

KESTEN'GA, Russian FederationWMO: **224030**Lat: **65.883N**Lon: **31.833E**Elev: **129**StdP: **99.78**Time Zone: **3.00 (E03)**Period: **82-90**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-27.0	-32.4	0.2	-30.0	-29.2	0.3	-26.9	8.7	-2.8	7.6	-3.1	0.8	330	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	23.9	16.7	21.8	15.8	20.1	14.7	17.6	22.4	16.4	20.7	15.4	19.3	2.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.4	19.8	14.6	10.5	18.2	13.5	9.8	16.9	50.1	22.7	46.4	20.7	43.5	19.6	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.3	5.5	-32.4	26.5	3.6	1.9	-35.0	27.9	-37.1	29.0	-39.2	30.1	-41.8	31.5		
(5)			-32.5	19.3	3.6	0.9	-35.1	20.0	-37.2	20.5	-39.2	21.1	-41.8	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-14.3	-11.6	-5.9	-0.9	5.8	11.6	14.6	11.5	7.4	2.2	-4.6	-11.6
	DBStd	10.97	8.37	7.07	5.09	4.31	3.94	4.72	3.20	3.76	3.42	3.98	5.99	7.26
	HDD10.0	3826	752	606	492	328	141	37	3	28	91	242	438	668
	HDD18.3	6549	1011	839	751	578	388	204	122	214	328	500	688	927
	CDD10.0	331	0	0	0	0	11	87	146	74	13	0	0	0
	CDD18.3	12	0	0	0	0	0	4	6	2	0	0	0	0
	CDH23.3	66	0	0	0	0	1	23	37	6	0	0	0	0
	CDH26.7	4	0	0	0	0	0	3	1	0	0	0	0	0
Wind	WSAvg	2.6	2.4	2.6	2.5	2.6	2.6	2.5	2.4	2.6	2.6	3.2	3.0	2.5
Precipitation	PrecAvg	570	38	31	29	28	49	58	77	66	56	49	48	42
	PrecMax	772	62	66	52	55	79	116	191	117	98	115	91	72
	PrecMin	438	8	8	15	8	15	24	46	25	10	16	5	13
	PrecStd	86	12	13	9	13	20	23	28	23	19	24	19	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	2.4	3.2	11.4	21.2	25.9	26.1	23.8	18.2	10.2	6.1	3.2
		MCWB	0.4	0.9	0.3	7.4	13.2	16.6	17.3	17.8	13.3	8.5	4.8	2.2
	2%	DB	0.3	1.1	2.0	8.8	17.7	23.4	24.4	21.5	15.1	9.4	4.5	1.5
		MCWB	-0.7	0.1	0.4	5.3	11.8	16.0	17.0	16.4	12.1	7.9	3.7	0.8
	5%	DB	-1.1	-0.2	1.3	6.7	14.8	21.2	22.5	19.1	13.5	8.7	3.5	0.4
		MCWB	-1.8	-1.1	0.1	3.8	9.6	14.9	16.5	14.6	11.0	7.4	2.9	-0.2
	10%	DB	-2.7	-1.7	0.6	5.2	12.5	19.1	20.6	17.1	12.2	7.8	2.2	-0.9
		MCWB	-3.2	-2.4	-0.3	2.2	7.8	13.9	15.1	13.7	10.3	6.7	1.6	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.2	1.4	7.9	14.4	18.2	19.4	18.5	14.1	9.5	5.0	2.3
		MCDB	1.4	1.9	2.1	10.3	20.6	23.4	24.2	23.0	16.6	9.8	5.5	2.9
	2%	WB	-0.6	0.3	0.7	6.2	12.1	17.1	17.9	16.8	12.6	8.2	3.9	1.0
		MCDB	0.2	1.0	1.5	8.7	16.9	21.7	22.7	20.5	14.6	8.9	4.3	1.4
	5%	WB	-1.7	-1.0	0.3	3.9	10.1	15.7	16.7	15.4	11.4	7.6	2.9	-0.1
		MCDB	-1.0	-0.2	1.1	6.2	14.0	20.7	21.1	18.5	13.1	8.5	3.4	0.4
	10%	WB	-3.2	-2.3	-0.2	2.6	8.3	14.1	15.7	14.1	10.5	6.7	1.6	-1.5
		MCDB	-2.6	-1.6	0.7	4.6	11.7	18.6	19.9	16.4	11.9	7.8	2.1	-0.9
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	6.6	7.5	7.5	7.1	7.3	5.8	4.9	3.8	4.4	6.5
		MCDBR	7.1	5.5	5.3	8.5	11.6	10.0	10.3	8.3	7.1	3.6	4.5	6.0
		MCWBR	6.7	5.3	4.2	5.9	6.6	4.9	5.1	4.3	4.5	2.9	3.9	5.8
	5% WB	MCDBR	6.9	5.3	4.6	8.2	10.6	9.4	9.0	7.8	6.5	3.5	4.0	5.6
		MCWBR	6.6	5.2	3.8	5.9	6.5	5.1	5.1	4.6	4.5	2.9	3.7	5.5
Clear-Sky Solar Irradiance	taub	0.204	0.270	0.301	0.333	0.347	0.343	0.365	0.342	0.317	0.290	0.353	0.202	
	taud	1.930	2.097	2.158	2.294	2.444	2.488	2.449	2.487	2.533	2.360	2.661	1.954	
	Ebn at Noon	238	582	754	829	858	870	833	821	763	582	249	36	
	Edh at Noon	19	56	87	99	95	93	94	82	63	44	19	5	
All-Sky Solar Radiation	RadAvg	0.10	0.63	2.15	3.80	4.59	5.16	4.76	3.44	1.90	0.66	0.14	0.02	
	RadStd	0.01	0.11	0.19	0.30	0.42	0.37	0.54	0.33	0.20	0.09	0.02	0.00	